

Dual-Domain MMD Adaptation: Held-Out Test Set and Large-Target Follow-Up

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1 What this document adds

Two things not present in `experiment_summary_v6.tex` or `freeze_epoch_ablation.tex`: (1) a genuinely held-out real test set, and (2) a first look at `real_large` (rather than `real_small`) as the adaptation target from a `syn_xlarge` source, at both YOLOv10n and YOLOv10m scale.

1.1 The held-out test set

Every real-domain comparison in this study up to now, baselines included, has been scored on the same 548-image real validation set – which is also the set used for checkpoint selection (`best.pt`) and early stopping during training. That is a mild form of leakage: the val set is never trained on directly, but it does influence which epoch’s weights get kept.

`real_test_heldout` closes that gap. It is the official VisDrone2019-DET *test-dev* split (1610 images), which is a separate download from the train/val archive this project’s `real_small/real_large` both draw from (`scripts/build_domain_datasets.py`’s `DEFAULT_REAL_ROOT`). Filename-level comparison confirmed zero overlap with the train+val pool used by any run in this study. Labels were remapped from VisDrone’s raw 12-category scheme to this project’s person/vehicle scheme, bicycles and ignored-regions dropped, matching the `yolo_splits_no_bikes` convention already used for `real_small/real_large`. All 1610 images retained at least one box after the remap. Built non-destructively: images symlinked, nothing existing modified.

2 `real_small` target, re-checked on the held-out set

Model	person mAP50	person mAP50-95	vehicle mAP50	vehicle mAP50-95
baseline (<code>real_small</code> , solo)	0.433	0.178	0.713	0.463
MMD (regularized, <code>syn_xlarge</code> pretrain)	0.403	0.163	0.699	0.451

Table 1: `real_small_1920_testing-3` (solo, COCO-direct) vs. `syn_xlarge_yolo_n_source_real_small_target_reg-f` both evaluated on `real_test_heldout` (1610 images, never used in any train/val split in this study).

Solo wins on every column, slightly more decisively than on the shared `real_large` val set used elsewhere in this study – ruling out shared-val checkpoint-selection leakage as the explanation for that earlier result.

3 `real_large` as the adaptation target (new)

Source = `syn_xlarge` (YOLOv10n pretrain, 150 epochs, imgsz 1920), target = `real_large` (6470 train images, vs. `real_small`’s ~ 1600) – the first time this study has used `real_large` rather than `real_small`

as the MMD target. Two MMD variants are compared against a solo (COCO-direct, no synthetic detour) baseline trained on the same target set.

Model	person mAP50	person mAP50-95	vehicle mAP50	vehicle mAP50-95
real_large_1920 (solo, COCO-direct)	0.703	0.345	0.907	0.451
syn_xlarge_source_real_large_target_reg_fr5 (freeze=6)	0.677	0.332	0.903	0.449
_stronger_weight-5 (120ep, freeze=25)	0.699	0.340	0.906	0.453

Table 2: Evaluated on `real_large` val (548 images).

Model	person mAP50	person mAP50-95	vehicle mAP50	vehicle mAP50-95
real_large_1920 (solo, COCO-direct)	0.482	0.204	0.751	0.341
syn_xlarge_source_real_large_target_reg_fr5 (freeze=6)	0.461	0.196	0.734	0.334
_stronger_weight-5 (120ep, freeze=25)	0.469	0.197	0.743	0.338

Table 3: Evaluated on `real_test_heldout` (1610 images, unseen). `real_large` as MMD target, source `syn_xlarge` (YOLOv10n). Solo wins on every column on both eval sets against both MMD variants, but the stronger-weight variant narrows the gap to the tightest margin seen anywhere in this study (0.003–0.007 mAP50-95 on the all-class average).

The matched no-MMD control for this regime (plain fine-tune of the `syn_xlarge` checkpoint on `real_large`, no MMD term) has not yet been run. Without it, the gap above conflates two variables – the synthetic-pretrain detour itself, and the MMD term – and cannot yet be attributed to MMD specifically. This is the natural next run to close out this section.

3.1 Model scale: YOLOv10m

Eval set	person mAP50	person mAP50-95	vehicle mAP50	vehicle mAP50-95
real_large	0.679	0.324	0.894	0.647
real_test_heldout	0.446	0.186	0.726	0.479

Table 4: `syn_xlarge_source_real_small_target` at YOLOv10m scale (source = `syn_xlarge_yolo_m-5`, target = `real_small`). Beats every YOLOv10n variant in this study on both eval sets, as expected for the larger backbone. No YOLOv10m solo baseline exists yet to isolate MMD’s contribution at this scale.

4 The other direction still holds: real→syn

Model	person mAP50	person mAP50-95	vehicle mAP50	vehicle mAP50-95
syn_small_1920_testing-2 (solo)	0.667	0.332	0.684	0.341
real_large_source_syn_small_target_reg-4 (MMD)	0.678	0.344	0.691	0.348

Table 5: Source = `real_large_1920`, target = `syn_small`, evaluated on `syn_large` val (640 images). This is the only comparison in this document where MMD beats solo on every column – consistent with `experiment_summary_v1`/README already flagging real→syn as the direction where adaptation genuinely helps, unlike syn→real.

5 Diagnostic: bandwidth-freeze does not prevent late-training drift

Investigated why `train/mmd_distance` climbs back up after the bandwidth freeze, across two runs with different freeze epochs:

- **freeze=6**: distance drops fast ($177,446 \rightarrow 157$) through epoch 6, then climbs straight back to $\sim 25,000$ by epoch 11 – superficially consistent with freezing mid-collapse, before the EMA had stabilized.
- **freeze=25**: distance was already low and fluctuating (18–1268) for several epochs *before* freezing, i.e. genuinely stabilized – yet still climbed steadily afterward (256 at epoch 27 $\rightarrow$ 7870 by epoch 45, roughly 300–400/epoch). This rules out “froze too early” as the explanation.

The behavior looks structural rather than a tunable timing issue: once frozen, the bandwidth stops tracking current feature geometry, and the MMD weight is simultaneously decaying by design (to avoid the collapse feedback loop documented in `experiment_summary_v2.tex`). Once detection loss dominates enough, nothing anchors the backbone’s features to where they were at freeze time, so raw distance from that now-stale reference drifts upward for the remainder of training regardless of which epoch the freeze happened at. Detection mAP kept improving normally in both affected runs, so this reads as an expected loss of signal in a late-training diagnostic rather than a training pathology worth chasing further via freeze-epoch tuning.

6 Open items

- No-MMD control for `real_large` target (source `syn_xlarge`, plain fine-tune, no MMD term) – needed to attribute the `real_large`-target gap to MMD specifically rather than the synthetic-pretrain detour alone.
- No YOLOv10m solo baseline on `real_small` or `real_large` – needed to check whether the solo-beats-MMD pattern holds at this model scale too.
- `real_test_heldout` has only been used with real-domain, `real_small/real_large`-target checkpoints so far; no equivalent held-out set exists yet for the synthetic domain.